MILITARY SPECIFICATION

BURSTER, INCENDIARY, FIELD,

M4 (W/O FUZE)

1. SCOPE

1.1 This specification covers one type of burster.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

O-A-51	-	Acetone, Technical.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-C-843	-	Cushioning, Material, Cellulostic.
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-P-704	-	Pail, Shipping, Steel (1 Through 12 Gallons).

MILITARY

MIL-G-155	-	Graphite, Dry (For Use in Ammunition).
MIL-C-3955	-	Cans, Fiber, Spirally Wound.
MIL-T-43036	-	Tape, Pressure Sensitive Adhesive, Plastic Film, Filament Reinforced (For Sealing Fiber Container and Cans).

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

DL37-4-54 - Burster, Incendiary, Field, M4 (W/O Fuze).
IDL37-4-54 - Burster, Incendiary, Field, M4 (W/O Fuze) - Inspection Data List.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing DL37-4-54 or on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the burster shall conform to the specifications and drawings listed on Drawing DL37-4-54 and subsidiary drawings.

3.2 Assembly. The burster shall be assembled in accordance with Drawing D37-4-54 (see 6.3).

3.3 Pellets.

3.3.1 Density. The pellets conforming to Drawing A37-4-28 shall have a density of no less than 1.55 nor more than 1.60 grams per cubic centimeter when tested as specified in 4.4.4.1(a).

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3.3.2 Forming. The pellets shall be consolidated from tetryl as specified on Drawing A37-4-28. Graphite conforming to grade I, II, III or IV of MIL-G-155 shall be uniformily blended with the tetryl as a lubricant and binder to facilitate manufacture. The quantity of graphite shall not exceed 2 percent when tested as specified in 4.4.4.1(b).

3.4 Ignition mixture. The burster (Drawing D37-4-54) shall contain a full column of ignition mixture conforming to Drawing B143-8-2, and weighing 230 ± 3 grams. The blended, granulated ignition mixture shall be dried to a constant weight (see 6.5) prior to filling the burster. While filling, the burster shall be tapped or otherwise vibrated to insure complete filling.

3.5 Silastic. The silastic, silicone rubber shall be applied uniformily around the tube end closures (Drawing D37-4-54) and cured for a minimum of 8 hours at room temperature prior to filling and after sealing the ignition mixture before assembly of the steel end cap. The interior surface of the steel tube in contact with the silastic shall be clean and dry before applying the silastic.

3.6 Leakage. The metal parts shall not leak when tested as specified in 4.4.4.2.

3.7 Functioning. The burster shall exhibit tube fragmentation and fuel ignition when tested as specified in 4.4.4.3 and 4.4.4.4.

3.8 Preproduction. Prior to the start of regular production a preproduction lot of bursters shall be produced in accordance with this specification for examination and tests (see 4.3).

3.9 Workmanship. The burster shall be free of foreign matter (oil, grease or dirt) and damage such as cracks, dents, bends or deformation which could affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.2 Government's responsibility. The Government will be responsible for the performance of the tests in 4.4.4.3 and 4.4.4.4. Samples shall be forwarded to the laboratory designated by the contracting officer (see 6.2).

4.1.3 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1, 3.4 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections.

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Lot. A preproduction lot of 100 bursters shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production.

4.3.2 Sampling.

4.3.2.1 For examination and nondestructive test. The entire preproduction lot shall be taken for examination and nondestructive test.

4.3.2.2 For destructive tests. The preproduction lot shall be randomly sampled in accordance with table I.

4.3.3 Inspection procedure.

4.3.3.1 For examination and nondestructive test. The preproduction lot shall be examined and tested in accordance with the classification of defects (4.4.3.3). The acceptance number is zero.

4.3.3.2 For destructive tests. The sample bursters shall be inspected and tested in accordance with table I.

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Table I. Sampling, inspection and acceptance/rejection criteria

Sample	Requirement	Test	Attributes	Accept	Reject
14	3.6	4.4.4.3*	Failure to detonate	0	1
			Failure to ignite fuel	0	1
6	3.6	4.4.4.4*	Failure to detonate	0	1
			Failure to ignite fuel	0	1

*Failure of squib to fire shall not reject the burster. Insert new squib and retest.

4.3.4 Acceptance/rejection criteria. The preproduction lot shall comply with the examinations and test specified in 4.3.3.1 to be acceptable. The acceptance number is zero. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the bursters of one interfix number, loaded by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions. However, no more than one lot of pellets or one lot of ignition mixture shall be represented in any one lot of finished bursters.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive test. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive tests.

(a) Chemical analysis. Sampling shall be conducted in accordance with MIL-STD-105, level S-4.

(b) Bursters. Sampling shall be conducted in accordance with MIL-STD-105, level S-3.

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4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive test. The sample bursters and the level A preparation for delivery shall be examined and tested in accordance with the classification of defects and MIL-STD-105.

4.4.3.2 For destructive tests.

(a) Chemical analysis. The sample pellets shall be tested in accordance with 4.4.4.1 and MIL-STD-105 using and AQL of 1.0 percent defective.

(b) Bursters. The sample bursters shall be tested in accordance with 4.4.4.3, 4.4.4.4 and MIL-STD-105 using and AQL of 1.0 percent defective.

4.4.3.3 Classification of defects. (See inspection data list IDL37-4-54 for inspection equipment required to perform the examinations specified in this section.)

(a) Bursters, incendiary, field, M4 (w/o fuze (Drawing D37-4-54).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 1.0 percent defective	
101	Component missing or incorrectly assembled	
102	Filling incorrect (3.4)	
103	Sealing incorrect (3.5)	
104	Inner assembly not sealed	
105	Outside diameter incorrect (Drawing B37-4-61)	
106	Marking missing or incorrect	
107	Workmanship (3.9)	
	AQL 0.65 percent defective	
108	Leakage	4.4.4.2
109	Pin incorrect (Drawing D37-4-55)	

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4.4.3.3 Classification of defects (continued).

(a) Burster, incendiary, field, M4 (w/o fuze) (Drawing C37-4-54) (continued).

Categories	Defects	Acceptance standards
Minor:	AQL 2.5 percent defective	
201	Threads incorrect (Drawing C37-4-29 & C37-4-64)	
202	Protective finish missing or incorrect	

(b) Preparation for delivery (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 4.0 percent defective
101	Container incorrect
102	Filler pads missing
103	Tape missing
104	Quantity incorrect
105	Blasting cap adapters, washers or gaskets missing
106	Container closure incorrect
107	Container damaged (broken or wracked)
108	Strapping missing or loose
109	Marking illegible or incorrect

4.4.4 Tests. (See inspection data list IDL37-4-54 for inspection equipment required to perform the tests specified in this section.)

4.4.4.1 Pellets.

(a) Density. Two calibrated aqeous zinc chloride solutions shall be prepared with density limits as specified in 3.3.1. The pellet, when subjected to the two solutions, shall submerge in the one of lower density and float in the one of higher density.

(b) Lubricant. A composite sample of approximately 10 grams (g) of tetryl, used for the density test, shall be weighed to the nearest 0.5 milligram (mg) in an approximate tared extraction thimble. An extraction apparatus containing approximately 50 milliliters (ml) of acetone

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conforming to O-A-51 shall be used. Heat shall be applied to the apparatus until a constant rate of reflux is reached. The minimum reflux rate shall be 60 drops per minute. The maximum reflux rate shall be such that the liquid level remains a minimum of 16 millimeters (mm) from the top of the thimble. The amount of solvent in the apparatus shall always be sufficient to keep the tetryl in solution. The extraction shall continue for a minimum period of 2 hours and for 1 hour after the filtrate becomes colorless. Following completion of the extraction, the thimbles shall be placed in a well-ventilated drying chamber, dried to a constant weight, at a temperature of 70° ± 2° C and the residue calculated as the percent of lubricant and inert material in the pellets.

4.4.4.2 Leakage. Prior to loading, the tube Drawing D37-4-55, the end cap, Drawing C37-4-56 and the plug, Drawing C37-4-29 shall be completely assembled including "O" rings and subjected to an internal air pressure of 3 to 5 pounds per square inch gage (psig) by connecting the plug to a regulated source of air. This assembly shall be immersed in water at a temperature of 0° to 5° F above the temperature of the assembly for a minimum period of 15 seconds during which time it shall be observed for leakage. Air bubbles from the assembly constitutes leakage and shall be cause for rejection. Acceptable assemblies shall be thoroughly dried prior to loading.

4.4.4.3 Incendiary burst. Each of the test bursters shall be employed in a simulated mine consisting of a 5-gallon can conforming to PPP-P-704, type I containing napalm (see 6.4). Each burster shall be armed with an M6 electric blasting cap (see 6.6) by means of an adapter conforming to Drawing C37-4-64 and fitted to the screw cap of the can with a support washer Drawing B37-4-65 and gasket Drawing B37-4-66 to reinforce the can-cap in a manner to prevent fuel leakage from the can or water seepage into the can or burster when the device is fully armed. Each can shall be placed with the cap corner down in a v-trench dug to a depth allowing for a minimum 2 inch cover of earth over the can. The device shall be fired and observed for tube fragmentation and fuel ignition. The earth in the area of emplacement shall be sifted for evidence of tube fragmentation. Tubes which are merely bursted with incidental partial shattering shall not be considered as fragmented. Any burning of fuel shall constitute evidence of ignition (See 6.3b).

4.4.4.4 Series burster (sympathetic detonation). The bursters shall be assembled in series, lengths of three bursters each. Each assembly shall be armed with an M6 electric blasting cap (see 6.6) by means of an adapter, washer and gasket (see 4.4.4.3) in a manner to prevent water seepage into the burster. Each armed assembly shall be firmly positioned at the bottom of a 6 inch deep v-trench into which a sealed 55-gallon drum of napalm is rolled in a manner to contact the bursters. The device shall

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be fired and the earth in the area of emplacement shall be sifted for evidence of tube fragmentation. Tubes which are merely bursted with incidental partial shattering shall not be considered as fragmented. Any burning of fuel shall constitute evidence of burning. If one of the bursters does not fire, the assembly shall be considered a defective (See 6.3c).

5. PREPARATION FOR DELIVERY

5.1 Level A packaging. The bursters shall be individually packaged in snug-fitting type II, grade B, style a, class 1 fiber cans conforming the MIL-C-3955. The container shall have approximate inside dimensions of 1-25/32 inches in diameter by 12-1/2 inches in length. Circular fiberboard filler pads, conforming to type CF, class-weather resistant, variety SW, PPP-F-320, shall be cut to such diameter so that pads will fit snugly inside container. One pad shall be placed on inside bottom of empty container, an after the burster has been placed therein, another pad shall be placed inside the can on top of burster. Additional pads shall be used, if required, to make a tight package. The can shall then be closed and the cover secured by sealing closure seam with minimum 1-1/2 inch wide pressure-sensitive tape conforming to MIL-T-43036. The tape shall have an overlap of approximately 3 inches and shall have the end folded back with adhesive to adhesive to form no less than 1 inch tab for easy removal of the tape. In addition, blasting cap adapters shall be individually wrapped using cellulostic cushioning material conforming to type II, class A or B of PPP-C-843; twenty support washers and 20 burster gaskets shall be wrapped in groups of 5 each using the same cushioning material.

5.2 Packing.

5.2.1 Level A. Twenty bursters packaged as specified in 5.1 shall be packed in an overseas type or class box conforming to PPP-B-621. The inside dimensions of the box shall be approximately 11-5/8 by 8-3/4 by 13 inches. In addition 20 blasting cap adapters, 20 support washers and 20 burster gaskets packaged as specified in 5.1, shall be packed between the packaged bursters in any convenient manner using additional cellulostic material to insure a tight pack. The top, bottom, sides and ends of each box shall be fitted with fiberboard filler pads fabricated from type CF, class-weather resistant, variety SW conforming to PPP-F-320. Additional pads shall be used, as required, to make a tight pack. The box shall be closed and strapped with galvanized strapping in conformance with the appendix of PPP-B-621.

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5.2.2 Level B. Twenty bursters, 20 blasting cap adapters, 20 support washers and 20 burster gaskets, packaged as specified in 5.1, shall be packed as specified in 5.2.1, except that coated finish strapping shall be used on domestic type of class containers.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages shall be marked in accordance with MIL-STD-129. Shipping containers shall be marked in accordance with Dwg D37-4-58.

5.4 Palletization. When specified in the contract or order, bursters packed as specified in 5.2.1 and 5.2.2 shall be palletized in accordance with Drawing C37-4-59. When the quantity of boxed bursters is insufficient to constitute a full pallet load, the boxed bursters shall not be palletized.

6. NOTES

6.1 Intended use. The M4 burster is intended for use with field improvised incendiary munitions filled with thickened fuel. It's purpose is to burst the fuel container, ignite the thickened fuel and scatter it over a large area. (See Department of the Army TB3-300-2).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Level of packing required.
(c) Preproduction details:

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether or not to proceed with production.
(4) Laboratory to which sample will be sent.

6.3 Caution.

(a) Due to the hazardous nature of filling the item, suppliers are advised to take all necessary precautions to protect personnel and prevent fires and explosions while assembling and loading the burster.

(b) Danger zone for 5-gallon cans is approximately 75 yard radius.

(c) Danger zone for 55-gallon drum using 3 bursters is eliptical approximately 450 yards long by 200 yards wide.

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6.4 Napalm formulation. For every 100-gallons of gasoline the M4 thickener and peptizer should be mixed with the gasoline in the following proportions: (See Department of the Army Technical Manual, TM3-366- Flame Fuels.)

Temperature	Thickner	Peptizer
Above 40° F	19.8 pounds	0
0° to 40° F	24.2 pounds	6 pounds
Minus 40° to 0° F	26.4 pounds	12 pounds

The thickener and peptizer should be added to the fuel in a large container (e.g., 55 gallon drum) and stirred into the gasoline with a wooden paddle until a gel has formed. The gel should then be stirred intermittently until an applesauce consistency results. After aging for six hours, the thickened fuel should be transferred to required containers leaving 2 inch void. Caution: Thickened fuel is subject to break down with moisture.

6.5 Ignition mixture. The final ignition mixture (Drawing B143-8-2) should be thoroughly dried prior to use. Acetone liquid must not be permitted to contact tetryl since swelling of pellets may result.

6.6 Blasting cap adapter. To arm the burster, the lead wires of the blasting cap are passed through the longitudinal slit in the adapter, and the wired end of the cap is then drawn to seat in the adapter bore. The gasket and support washer are assembled onto the adapter. The assembly is inserted through a 13/32 inch hole in the screw cap of a five gallon can or container closure to be employed and tightened securely in place in the burster plug. An M8 blasting cap may be used, its use however, shall be reported in the test data. Dow Corning Silastic, 732 RTV, silicone rubber will resist fuels as well as water and may be employed to assure sealing of the armed burster. The M6 and M8 blasting caps may be procured from the following:

E.I. du Pont de Nemours and Company - E1A(6), E1B(6)
E1A(8), E1B(8)
Hercules, Incorporated - Cap, Blasting, Special Electric, M6.
Cap, Blasting, Special Electric, M8.

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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